



A cooperative optimization strategy for distributed multi-robot manipulation with obstacle avoidance and internal performance maximization

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ABSTRACT

Reactive multi-robot manipulation can be of great assistance in many applications. In this article a distributed cooperation strategy for object manipulation with multiple robots in un-modelled environments is proposed to tackle workpiece transportation and dynamic obstacle avoidance simultaneously. A decentralised optimisation process will run first to generate robot pose references and then a joint-level control is responsible for tracking. Since in multi-arm manipulation systems the available workspace for each robot is more restricted, necessary internal performance optimisation is also included in the planning phase. A velocity-level optimum will be solved and then transformed to a position-level control reference so that it can be combined with many compliant interactive controllers such as impedance control. Although no torque or force sensor is required and each robot only has access to local information, when avoiding obstacles the coordination strategy can generate synchronised group motion so that internal forces are avoided. Only task-space variables will be communicated, therefore it is easy to team up different types of robots, making the system more flexible. Simulation and experiments with two redundant manipulators have been done to validate the proposed design, and data show that the online computation speed is fast.

1. Introduction

1.1. Background

In modern industry, robots have become indispensable due to its accuracy and repeatability. And because industrial tasks with increasingly high complexity are desired to be performed by robots instead of human, the interest in multi-robot systems has been growing in recent decades [1]. Such systems are able to handle larger workloads, accomplish more sophisticated tasks, and they can be more flexible by either teaming up to work on a large single task or ungrouping for smaller parallel tasks depending on demand. Furthermore, with the development of “Cobots” and the concept of robotic society [2], in more and more scenarios are robots desired to work in dynamic environments with human. Therefore, the safe and intelligent control of multi-robot systems under environment uncertainties is strongly required. This article will focus on the cooperative robot control for manipulation tasks, where several robots are desired to jointly move a rigid workpiece to a desired pose. Note that some other works achieve similar goals with only mobile vehicles [3,4], but the Degree Of Freedom (DOF) is often limited. In this work the systems with dexterous robot arms will be discussed, and the method can be extended to mobile manipulators.

The primary aim for the manipulation system is to precisely move the object to the desired Cartesian pose. At the same time, the robots should be able to detect and react to unpredictable obstacles online, because the realistic environment is often dynamic and not fully modelled, especially when human and robots co-exist without isolation. To be specific, online collision avoidance for both the object and robot bodies is a necessity in this case, ideally with minimal influence on workpiece transportation accuracy. This is however not straightforward for cooperative manipulators, since the inter-robot physical connection introduced by the grasp of a single object transfers the kinematic and dynamic constraints of all cooperating agents to each individual robot, greatly reducing its free operation room and limiting the capability of adjustment motion, making it essential to pursue maximum agility and performance during cooperation by carefully tuning the robot pose based on kinematic criteria like mechanical joint limit [5] and maximum Cartesian workspace [6], and dynamic criteria like manipulability [7], joint torque and speed limitation, and so on. Moreover, due to differences in sensing range and joint-space performance condition (e.g. when one robot detects an obstacle while the others do not), the robot members may make different motion decisions that lead to asynchronisation of the intended workpiece motion, which can generate large and even dangerous internal conflicting forces on a rigid

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object [8]. This effect is less frequently investigated because for soft or no physical connection it is seldom critical [9,10], but in this paper it will be treated as an important performance aspect. In fact, hardly any work can be found dealing all the aforementioned specific issues concurrently.

It is then obvious that cooperative manipulation control in dynamic environments is a multi-objective problem, therefore optimisation techniques have become a natural choice to achieve the goal. Many existing researches seek to optimise a complete motion trajectory [11–13], but these motion planners only consider very limited optimisation objectives and they are too computationally demanding to achieve fast online reaction. In addition, the un-modelled variation of the environment could also make the predicted optimal trajectory invalid soon. In this work only the instantaneously optimal robot pose will be planned so that the control program can run at a higher frequency. One common and important challenge here is the nonlinearity in both robot kinematics and dynamics, which often imposes the employment of rigorously derived numerical methods [14] or even learning and neural network approaches [15]. The authors of this article will, on the other hand, propose a novel and more physically interpretable idea to solve the optimisation problem efficiently by first obtaining an analytical velocity-level solution and then transforming it to the position level. More importantly, internal performance criteria will be included in the cost functions.

For multi-robot systems with arms, the high dimension is also challenging considering the limitation of computational power in reality. Although centralised controllers are often adequate for smaller robot groups since they enable easier synchronisation between different robots [9,16] and allow direct adoption of single-robot regulating tools like inverse kinematic control [17], inverse dynamic control [18], and grasping control with all manipulators modelled as fingers of a single hand [19], as the number of cooperating agents increases it will be difficult for the central computer to catch up. This work endeavours to develop a highly scalable framework, hence a decentralised structure is preferred. Early attempts have introduced the leader–follower type of control [20], where a master role is assigned to one robot who is responsible for sensing the environment and making decisions for the whole group, while the other members receive and follow a consensus from the leader in order to realise coordination. This structure however has single point of failure and the scalability is still limited by the master robot hardware. To achieve full decentralisation, the manipulation motion task can be provided *a priori* to all robots so that each of them can be controlled separately to track the desired trajectory even without communication [21–23], but in presence of local disturbances (e.g. obstacle avoidance) coordination could be violated due to the lack of information sharing and thereby internal load may arise [24].

For these reasons communication between robots will be established in this work, making the overall control structure distributed. This kind of topology is becoming popular for multi-robot systems, however to the best knowledge of the authors, most of its implementational attempts have been applied to collaborative mobile platforms [10,25]. A few experimentally validated works with actuated arms do exist recently. For example, [26] presents a fully distributed adaptive control for cooperative manipulation with uncertain parameters, and the algorithm is implemented on three redundant mobile manipulators during experiments. Another research [27] by the same institute focuses on the integration of event-triggered communication into the distributed structure, with experimental validation performed on a dual-arm robot. While they have worked on the theoretical completeness nicely, none of them is dealing with the issues resulting from dynamic obstacles. This article proposes to fill this gap. In the meantime, the minimal amount of transferred data and the eluding of exchanging joint-space information will be pursued, so that the system would be more plug-and-play-friendly when the cooperative group consists of heterogeneous robot types.

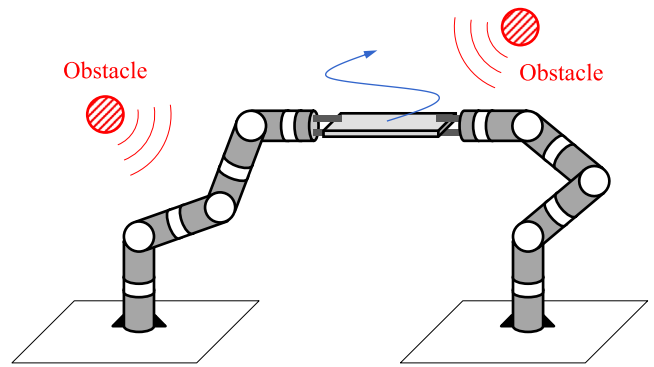


Fig. 1. The cooperative robot manipulation scenario considered in this paper.

1.2. Contribution of this work

This paper is an extended work of [28], which presents the basic framework of the cooperative manipulation control for multi-robot system in a dynamic environment, as shown in Fig. 1. Compared with aforementioned related works, this paper has the following contributions

1. A multi-robot cooperation strategy for rigid object manipulation with simultaneous whole-body obstacle avoidance and internal performance maximisation is designed based on a multi-objective optimisation mechanism that includes the subtask criteria mentioned above, and the proposed approach is validated in experiments.
2. A novel computationally efficient iterative quadratic programming method with clear physical interpretation is developed for indirectly solving the nonlinear optimisation problem in a distributed framework. Experiments show an online computation time within 10 ms, even better than many motion planning algorithms for single robots.
3. A plug-and-play-friendly negotiation protocol is proposed to guarantee coordination between robots despite different local sights of the obstacles and joint-space performance conditions and that each robot only solves a local optimisation problem without centralised information. The negotiation only requires a limited amount of data exchange, enabling the application in unstable wireless networks and thus also the extension to mobile manipulators.

Compared with the preceding work [28], this article mainly adds the following extensions:

1. The internal optimisation motion is computed online based on real-time evaluation of an explicitly defined performance index;
2. Unnecessary object path tracking error is avoided by shifting internal optimisation to null space;
3. New experiments and simulation are designed and more detailed data analysis is presented;
4. The obstacle detection and data exchanging server hardware is substantially improved;

In the remaining part of this paper, the kinematics of the multi-robot system and the dynamics of individual robot arms will be shown in Section 2. Then in Section 3 the idea of solving the locally optimal pose based on iterative quadratic programming is introduced, and Section 4 will talk about the negotiation mechanism for achieving coordinated motion using the proposed optimisation method. The interactive manipulator control will be briefly discussed in Section 5. Simulation and experimental validation of the approach are shown in Sections 6 and 7, respectively. And finally Section 8 provides a summary of this whole manuscript.

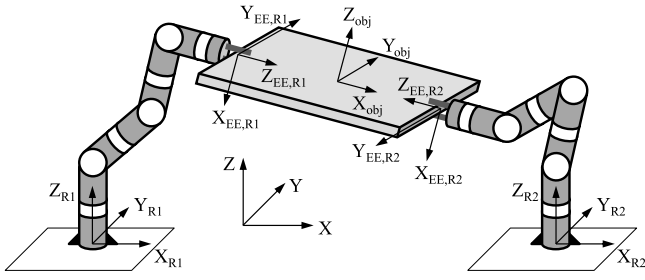


Fig. 2. The kinematics of the multi-robot manipulation system.

2. Description of the system

The scenario discussed in this paper is shown in Fig. 1. The manipulation system consists of multiple redundant robot arms grasping a single rigid object and the ultimate goal is to manipulate it towards a desired pose. Although only two manipulators are illustrated, the proposed methodology can be generalised for larger systems without difficulty. Instead of considering static environments, in this work the obstacles are dynamic and their motion is considered unpredictable. The following assumptions are made about the manipulation system:

1. The grasping is firm, i.e. there is no relative motion between the grasped object and each end effector. This is a very common type of grasping for fingered grippers that come with most robot products by default.
2. The grasping locations on the object are known to the robots. This assumption is natural because it is either that the workpiece information is provided by human in advance, where the grasping locations can be easily assigned, or that each robot recognises the object using optical sensors, during which it can select a grasping point on its own with a known position.
3. The robots are equipped with sensors (e.g. LiDAR, UWB, cameras, etc.) that can measure the positions of surrounding obstacles. This is especially valid in human-robot co-existence scenarios, where infrastructures are often pre-built in the working area and wearable markers or UWB tags can be placed on moving humans. There are also robot products with on-body cameras that detect objects in a mobile fashion.
4. A communication network is available within the robot group. For fixed-based robot systems, this can be established with wired buses for robustness, while for mobile manipulators a wireless network is preferable, e.g. with WiFi. This set-up is nowadays common in logistic centres and on production lines.

The coordinate frame system is depicted in Fig. 2. The object frame is denoted with the subscript “obj”. The end effector frame is denoted by “EE”. “Ri” indicates the *i*th robot, and it is used in the notation of the corresponding robot base frame. The notation of the world frame has no subscript.

In this manuscript, the joint variable vector will be denoted as q , the Cartesian twist (velocity) vector as $\dot{x}$. The 4-by-4 Cartesian pose matrix is denoted as

$$T = \begin{bmatrix} R & p \\ 0_{1 \times 3} & 1 \end{bmatrix} \quad (1)$$

with R being the orientation and p the position. Each kinematic variable may have a subscript for noting which coordinate frame it is describing. A left superscript will indicate in which frame the variable is represented.

The grasping geometry can be mathematically characterised by the grasping matrix, as explained in [8,29], which is defined for each grasping point i as

$$G_i = \begin{bmatrix} I_{3 \times 3} & 0 \\ S(\text{obj } p_{EE,Ri}) & I_{3 \times 3} \end{bmatrix} \quad (2)$$

where I is the identity matrix, and $\text{obj } p_{EE,Ri}$ is the position of the i th end effector represented in the object frame. $S(\cdot)$ is the corresponding skew symmetric matrix of its operand vector. With the aforementioned grasping assumptions, G_i is a known constant. If the wrench exerted by the i th end effector is denoted as h_i , the internal wrenches are calculated by

$$\text{obj } h_{\text{int}} = (I - G^\dagger G) \text{obj } h \quad (3)$$

where $\text{obj } h = [\text{obj } h_1^T \text{obj } h_2^T \dots \text{obj } h_{N_R}^T]^T$ is a $6N_R$ -by-1 aggregated wrench vector, and $G = [G_1 \ G_2 \dots G_{N_R}]$, with N_R being the total number of cooperating robots. $G^\dagger$ denotes the pseudo-inverse of G .

For each robot, the end effector pose represented in the robot base frame, ${}^{Ri}T_{EE,Ri}$, can be computed with forward kinematics depending on the joint angle q_{Ri} . According to the grasping assumption, ${}^{EE,Ri}T_{\text{obj}}$ is known, therefore the object pose T_{obj} can be further computed locally, and its Jacobian matrix w.r.t. Robot i , $J_{\text{obj},Ri}$, can also be obtained using standard procedures.

The dynamic equation of the i th robot arm is

$$M_{Ri} \cdot \ddot{q}_{Ri} + C_{Ri} \cdot \dot{q}_{Ri} + g_{Ri} = \tau_{m,Ri} + \tau_{f,Ri} + \tau_{e,Ri} \quad (4)$$

where the matrix M is the inertia matrix, C contains the centrifugal and Coriolis terms, and g is the gravity torque vector. τ_m is the control input torque. τ_f is the friction torque. τ_e is the external torque, which according to the principle of virtual work has the following property

$$\tau_{e,Ri} = -J_{EE,Ri}^T \cdot {}^{Ri}h_i \quad (5)$$

3. Local optimal pose reference

The goal of simultaneously achieving transportation, obstacle avoidance and joint-space pose tuning can be naturally formulated as an optimisation problem, however the direct solution in the robot joint space is nonlinear and often high dimensional for agile redundant manipulators. Therefore this chapter presents a novel and physically interpretable idea that first formulates and solves a velocity-level quadratic programming problem with analytically derived solution, and then transforms it into position level so that the outcome becomes compatible with popular compliant motion controllers.

Note that due to the changefulness of the dynamic environment only the instantaneous optimal pose will be solved. Moreover, in the proposed framework each robot first optimises only locally and then the global goal is achieved by a coordination protocol introduced in Section 4, so the treatment presented in this section is carried out w.r.t. each individual manipulator, and all kinematic variables are assumed to have already been transformed into the local robot base frame, but for better readability the “Ri” super-/subscripts will be omitted.

3.1. Velocity-level solution

While object transportation is the primary goal of the multi-robot manipulation system, when an obstacle appears unexpectedly in the way it is necessary to sacrifice some tracking performance for safety, therefore the cost function is designed to contain a goal tracking part and a collision avoidance part. To achieve whole-body reaction to obstacles, the latter part will further include two components — one for the grasped object and the other for the robot arm. The workpiece will be driven as far away as possible from the online detected obstacles according to their distances. And by defining N_j key points on the robot links and attend to each of them, the same principle can be applied to secure the entire robot arm. To minimise transportation error, the collision avoidance parts are deactivated with a zero weight when the obstacles are outside a pre-defined reaction radius. In [28] internal pose optimisation also has an individual cost component based on joint limits, manipulability, etc., but that would lead to persistent transportation error, so in this paper it will be treated differently. The

quadratic programming problem is formulated in joint velocity level as follows to facilitate closed-form analytical solution:

$$\begin{aligned} \min_{\dot{q}_{d0}} g_0(\dot{q}_{d0}) = & \frac{1}{2} (\dot{q}_{d0} - \dot{q}_{trp})^T \mathbf{W}_1 (\dot{q}_{d0} - \dot{q}_{trp}) \\ & + \sum_{l=1}^{N_{obs}} \frac{1}{2} (\mathbf{v}_{obj,d0} - \mathbf{v}_{obj,obs,l})^T \mathbf{W}_{2,l} (\mathbf{v}_{obj,d0} - \mathbf{v}_{obj,obs,l}) \\ & + \sum_{j=1}^{N_j} \sum_{l=1}^{N_{obs}} \frac{1}{2} (\mathbf{v}_{j,d0} - \mathbf{v}_{j,obs,l})^T \mathbf{W}_{3,jl} (\mathbf{v}_{j,d0} - \mathbf{v}_{j,obs,l}) \end{aligned} \quad (6)$$

with $\dot{q}_{d0}$ being the desired optimal joint velocity. Next, the three terms of the cost will be explained one by one in detail.

The first cost component is for object transportation, where the diagonal matrix $\mathbf{W}_1 = w_{trp} \mathbf{I}$ is the corresponding weight and $\dot{q}_{trp}$ is a reference joint velocity for the transportation task. It also includes a null space control part in order to incorporate internal performance optimisation, i.e.,

$$\dot{q}_{trp} = \mathbf{J}_{obj}^\dagger \dot{\mathbf{x}}_{trp} + (\mathbf{I} - \mathbf{J}_{obj}^\dagger \mathbf{J}_{obj}) \dot{q}_{ic} \quad (7)$$

where $\dot{\mathbf{x}}_{trp}$ is the desired goal tracking velocity of the workpiece in Cartesian space, and $\dot{q}_{ic}$ is a joint velocity for minimising an internal cost. One advantage here of the formulation (6) is that the reference velocities can be designed intuitively and flexibly based on physical meanings. For example, they can be determined by a PI law based on the pose error, or by artificial potential field method. Convergence is naturally achieved as long as the adopted approaches are individually stable. Since the main purpose of this manuscript is to demonstrate the overall framework, here $\dot{\mathbf{x}}_{trp}$ is determined only by a simple P control law:

$$\dot{\mathbf{x}}_{trp} = \mathbf{K}_{trp} \tilde{\mathbf{x}}_{obj,trp} \quad (8)$$

with $\tilde{\mathbf{x}}_{obj,trp}$ being the error between the actual object pose and the transportation goal. Note that the lower part of $\tilde{\mathbf{x}}_{obj,trp}$ should be computed using either axis-angle or unit quaternion representation. $\mathbf{K}_{trp}$ is a positive-definite diagonal gain matrix.

The internal performance cost will be denoted as g_{ic} . Its mathematical formulation varies with user requirement. The most widely considered index is the manipulability measure. Joint limits and torque efficiency are also of interest. To minimise g_{ic} , $\dot{q}_{ic}$ can be designed as

$$\dot{q}_{ic} = -k_{ic} \left(\frac{\partial g_{ic}(\mathbf{q})}{\partial \mathbf{q}} \right)^T \quad (9)$$

with k_{ic} being a positive gain. If an analytical expression for g_{ic} is available, the gradient could also be derived symbolically and computed fast online, otherwise numerical methods have to be employed. However if g_{ic} depends completely on the robot pose, an alternative way is to first solve the following optimisation problem offline

$$q_{ic} = \operatorname{argmin}_{q_{ic}} g_{ic}(q_{ic}) \quad (10)$$

and compute $\dot{q}_{ic}$ online using

$$\dot{q}_{ic} = k_{ic} (q_{ic} - q) \quad (11)$$

The second component of the cost in (6) is related to obstacle avoidance for the grasped object. $\mathbf{v}_{obj,d0}$ is the translational velocity of the object, expressed by

$$\mathbf{v}_{obj,d0} = \mathbf{J}_{v,obj} \dot{q}_{d0} \quad (12)$$

where $\mathbf{J}_{v,obj}$ is the upper three rows of $\mathbf{J}_{obj}$. N_{obs} is the total number of detected obstacles, which are treated as points. To determine $\mathbf{v}_{obj,obs,l}$, a reaction radius r_{obs} is firstly defined. The idea is that only the obstacles that are close enough to the object will be processed. A repulsive velocity is assigned to $\mathbf{v}_{obj,obs,l}$ as

$$\mathbf{v}_{obj,obs,l} = \begin{cases} k_{obs} (\mathbf{p}_{obj} - \mathbf{p}_{obs,l}), & \text{if } \|\mathbf{p}_{obj} - \mathbf{p}_{obs,l}\| < r_{obs} \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (13)$$

where $\mathbf{p}_{obs,l}$ is the position of the l th obstacle point, and k_{obs} is a positive gain. As can be seen, again a P control is used to generate the reference velocity. The weighting matrix is determined as follows

$$\mathbf{W}_{2,l} = \begin{cases} w_{obs} \mathbf{I}, & \text{if } \|\mathbf{p}_{obj} - \mathbf{p}_{obs,l}\| < r_{obs} \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (14)$$

where the positive number w_{obs} is the nominal weight. The conditional assignment of zero is necessary because otherwise according to (13) an obstacle outside the reaction range will cause the object velocity to tend towards zero.

The purpose of the third term in (6) is obstacle avoidance for the robot arm. The principle is basically the same as that for the grasped object, but now multiple key points on the robot body must be chosen, and the computation will be done for each of them. Obviously the more key points there are, the better a successful avoidance will be guaranteed, but it also generates more computational burden. A simple strategy is to choose the joints, since their Jacobian matrices, which will be used in the optimisation, are usually intermediate results when computing the end effector Jacobian. N_j is the total number of the selected key points, and $\mathbf{v}_{j,d0}$ is equal to

$$\mathbf{v}_{j,d0} = \mathbf{J}_{v,j} \dot{q}_{d0} \quad (15)$$

where $\mathbf{J}_{v,j}$ is the upper part of the Jacobian matrix of the j th key point on the arm. The reference avoidance velocity $\mathbf{v}_{j,obs,l}$ and the weight $\mathbf{W}_{3,jl}$ are calculated using the same rules as (13) and (14).

Since the optimisation variable in (6) is the joint velocity, during the solution the position-level variables can be treated as constant coefficients. By letting the gradient of g_0 be zero, it can be straightforwardly shown that the optimal joint velocity is

$$\dot{q}_{d0} = \mathbf{A}^{-1} \mathbf{b} \quad (16)$$

where

$$\begin{aligned} \mathbf{A} = & \mathbf{W}_1 + \sum_{l=1}^{N_{obs}} \mathbf{J}_{v,obj}^T \mathbf{W}_{2,l} \mathbf{J}_{v,obj} + \sum_{j=1}^{N_j} \sum_{l=1}^{N_{obs}} \mathbf{J}_{v,j}^T \mathbf{W}_{3,jl} \mathbf{J}_{v,j} \\ \mathbf{b} = & \mathbf{W}_1 \dot{q}_{trp} + \sum_{l=1}^{N_{obs}} \mathbf{J}_{v,obj}^T \mathbf{W}_{2,l} \mathbf{v}_{obj,obs,l} + \sum_{j=1}^{N_j} \sum_{l=1}^{N_{obs}} \mathbf{J}_{v,j}^T \mathbf{W}_{3,jl} \mathbf{v}_{j,obs,l} \end{aligned} \quad (17)$$

Since $\mathbf{W}_1$ is positive-definite, $\mathbf{A}$ is always invertible. But the pseudo-inverse of Jacobian is used in the null space control (9), hence singularity still has to be handled carefully. An inclusion of manipulability in the internal cost function g_{ic} (will be shown in Section 6) can often effectively keep the robot pose away from singularities. If the singularity is caused by the end-effector pose and cannot be tackled by internal motion, e.g., the end effector has reached the maximum workspace boundary, damped least-square inversion can be utilised. Note that this treatment will modify the inversed matrix, hence there could be a negative effect on motion accuracy, which later induces imperfect coordination and some internal load. A quantitative study of this effect and the reasonable choice of the damping constant can be found in [30]. However, since this type of singularity is more easily interpretable in the Cartesian space, its possibility could also be lowered by shaping a well-conditioned path during transportation motion design phase.

3.2. Position-level solution

The analytical solution (16) in the previous section is a composition of transportation, internal optimisation and obstacle avoidance motion and it can be easily computed online. Although most industrial manipulators have a built-in velocity control mode, it usually produces rigid and aggressive interactive behaviour. Therefore, instead of directly feeding the optimal joint velocity reference to the low level controller, this article will present an approach to transform it into position level, since only in that way can compliant robot controllers like PD

torque control and impedance control be integrated for better safety in physical interaction

The $\mathbf{q}_{d0}$ as in (16) will be firstly multiplied by a small time step Δt to obtain a pose increment, which will be added to the actual pose $\mathbf{q}$ to get an updated reference pose $\mathbf{q}_{d0}$. This reference pose is usually not suitable for compliant control in practice, since $\dot{\mathbf{q}}_{d0}\Delta t$ can be so small that the resulting control torque command is not enough to overcome friction and the robot would barely move. Even if there is no un-compensated friction, due to inertia a small input torque could probably not generate remarkable motion within a short program cycle time. Therefore instead of being fed into control, this $\mathbf{q}_{d0}$ will be substituted into the optimisation problem (6) as the actual pose and the quadratic programming problem will be solved again. The resulting optimal velocity will be used to update $\mathbf{q}_{d0}$, and this process will be repeated. The iteration can be stopped either when the solution $\dot{\mathbf{q}}_{d0}$ becomes almost zero or when a maximum number of loops has been reached. Only the final $\mathbf{q}_{d0}$ is used as the control reference.

This iterative solving process is similar to some Jacobian-based inverse kinematic solvers, which can be understood as an online simulation with the robot dynamics modelled as an integrator. The initial state is the current joint pose, and the simulated duration T_{itr} is equal to the product of Δt and the maximum number of iterations. Instead of directly solving an optimal pose, the algorithm actually computes an optimal motion that drives the robot to an optimal pose, which is the $\mathbf{q}_{d0}$ at the end of the simulation. The basic idea is also similar to Sequential Quadratic Programming (SQP), however as long as the reference velocities of subtasks in Section 3.1 are physically interpretable, different planning laws can be used and the strict mathematical formulation in SQP is not necessary, making the method more flexible.

The analytical solution (16) and (17) can be computed fast online and therefore it is suitable for real-time responsive control. As will be shown later, only a few milliseconds are required for hundreds of iteration loops. The pseudocode of this approach is attached, labelled as Algorithm 1.

Algorithm 1 Single-robot optimisation

```

1: function LOCALOPTIM( $\mathbf{q}_{now}, \mathbf{x}_{d,trip}, \mathbf{p}_{obs}$ )
2:   Initialise  $\mathbf{q}_{d0} = \mathbf{q}_{now}$ ,  $t_{itr} = 0$ 
3:   repeat
4:     Compute the object pose based on  $\mathbf{q}_{d0}$ 
5:     Compute key point positions based on  $\mathbf{q}_{d0}$ 
6:     Compute all Jacobians based on  $\mathbf{q}_{d0}$ 
7:     Compute  $\dot{\mathbf{q}}_{trp}$ 
8:     Compute every  $\mathbf{v}_{obj,obs,l}$ ,  $\mathbf{W}_{2,l}$ ,  $\mathbf{v}_{j,obs,l}$ ,  $\mathbf{W}_{3,jl}$ 
9:      $\dot{\mathbf{q}}_{d0} := \mathbf{A}^{-1}\mathbf{b}$ 
10:     $\mathbf{q}_{d0} := \mathbf{q}_{d0} + \dot{\mathbf{q}}_{d0}\Delta t_{itr}$ 
11:     $t_{itr} := t_{itr} + \Delta t_{itr}$ 
12:  until  $t_{itr} \geq T_{itr}$  ▷ or until  $\dot{\mathbf{q}}_{d0} \& \ddot{\mathbf{q}}_{d0} \approx 0$ 
13:  return  $\mathbf{q}_{d0}$ 
14: end function

```

4. Coordination strategy

As mentioned before, the robots are only aware of its own surroundings, so a manipulator may react to an obstacle that the other members do not see yet. This will result in different intentions of object motion among the robot group and internal forces could arise. The coordination of distributed cooperative robot arms in the discussed multi-objective set-up has seldom been investigated so far, therefore a low-cost negotiation mechanism will be proposed to guarantee non-conflicting motion. The goal is to achieve consensus on the desired trajectory of the physically coupled part despite heterogeneous local optima resulting from the decentralised fashion, while as little information as possible is transferred during communication. Preferably no joint-space data should be exchanged, since otherwise the complete

models of all robots must be known to each other and the framework would be difficult to implement if various robot types exist in the cooperating group.

The basic idea of the negotiation is to determine a compromised optimum of all the local optima and solve an optimisation problem again to obtain the final control reference. The details are explained as below: First on each robot Algorithm 1 is still used to generate a locally optimal robot pose $\mathbf{q}_{d0,Ri}$, however it will not be provided to the motion control. A corresponding Cartesian pose of the object, denoted as $\mathbf{T}_{obj,d0,Ri}$, is computed based on $\mathbf{q}_{d0,Ri}$ with forward kinematics and then broadcast via the communication network together with a weight, whose determination will be talked about soon. Each robot will receive the locally optimal object pose from all robots, then the weighted average of them are computed, denoted as $\mathbf{T}_{obj,d}$. This mean Cartesian pose will be used as the globally desired object pose for the whole robot group. The weighted mean position is computed with

$$\mathbf{p}_{obj,d} = \sum_{i=1}^{N_R} w_{Ri} \cdot \mathbf{p}_{obj,d0,Ri} \quad (18)$$

where w_{Ri} is the normalised weight. The average orientation $\mathbf{R}_{obj,d}$ can be computed with the help of unit quaternion as introduced in [31], and then $\mathbf{T}_{obj,d}$ can be constructed.

The cost in the last iteration loop of Algorithm 1 is chosen to determine the weight w_{Ri} , because it indicates the overall performance at the simulated steady state. If it is high, it means that the virtual motion is somehow constrained and there is a non-zero trade-off between transportation and obstacle avoidance, making this robot deserve more consideration in cooperation. Each robot will receive the last-step cost $g_{0,T,Ri}$ from all members, and they will be normalised as weights

$$w_{Ri} = \frac{g_{0,T,Ri} + d}{\sum_{i=1}^{N_R} (g_{0,T,Ri} + d)} \quad (19)$$

where d is a damping factor to avoid division by zero.

The global goal pose $\mathbf{T}_{obj,d}$ will be the new reference pose for the object and every manipulator should stick to it. Obviously the locally desired optimum of one robot may no longer be fulfilled when the other robots are reacting to a situation that this robot cannot see, but the sacrifice in local performance is necessary in this case for eliminating conflict in end effector motion. This is a result of the constraint induced by the physical connection in the manipulation system.

Note that both the locally and globally desired object poses above are represented in the world frame. On each robot, for computation efficiency, the global optimal pose will be transformed into the local robot base frame before proceeding. In the remaining part of this section, the transformation is assumed to be already done, and again the “Ri” super-/subscripts are omitted for better readability.

To guarantee non-violated grasping formation, a second optimisation problem is formulated with the motion to $\mathbf{T}_{obj,d}$ as an equality constraint

$$\begin{aligned} \min_{\dot{\mathbf{q}}_d} \quad & g_0(\dot{\mathbf{q}}_d) = \frac{1}{2} (\dot{\mathbf{q}}_d - \dot{\mathbf{q}}'_{trp})^T \mathbf{W}_1 (\dot{\mathbf{q}}_d - \dot{\mathbf{q}}'_{trp}) \\ & + \sum_{j=1}^{N_j} \sum_{l=1}^{N_{obs}} \frac{1}{2} (\mathbf{v}_{j,d} - \mathbf{v}_{j,obs,l})^T \mathbf{W}_{3,jl} (\mathbf{v}_{j,d} - \mathbf{v}_{j,obs,l}) \end{aligned} \quad (20)$$

$$\text{s.t. } \mathbf{J}_{obj} \cdot \dot{\mathbf{q}}_d = \dot{\mathbf{x}}_{obj,d}$$

where

$$\dot{\mathbf{x}}_{obj,d} = \mathbf{K}_{glo} \bar{\mathbf{x}}_{obj,d} \quad (21)$$

and $\bar{\mathbf{x}}_{obj,d}$ is the pose error between $\mathbf{T}_{obj}$ and $\mathbf{T}_{obj,d}$. $\mathbf{K}_{glo}$ is a P control gain matrix. Compared with the first optimisation problem, here in the first cost component $\dot{\mathbf{q}}_{trp}$ is replaced with

$$\dot{\mathbf{q}}'_{trp} = \mathbf{J}_{obj}^\dagger \dot{\mathbf{x}}_{obj,d} + (\mathbf{I} - \mathbf{J}_{obj}^\dagger \mathbf{J}_{obj}) \dot{\mathbf{q}}_{ic} \quad (22)$$

because of the modified target pose. The second term in the cost of (6), i.e. the obstacle avoidance for the object, is removed since the object

motion is already fixed by the equality constraint. The other variables in the second quadratic programming problem (20) are determined in the same way as in Section 3.1. It can be shown with the method of Lagrangian multipliers that the solution to (6) is

$$\dot{q}_d = \Gamma^{-1}(\Theta + J_{\text{obj}}^T \lambda) \quad (23)$$

where

$$\Gamma = W_1 + \sum_{j=1}^{N_j} \sum_{l=1}^{N_{\text{obs}}} J_{v,j}^T W_{3,j,l} J_{v,j} \quad (24)$$

$$\Theta = W_1 \dot{q}'_{\text{trp}} + \sum_{j=1}^{N_j} \sum_{l=1}^{N_{\text{obs}}} J_{v,j}^T W_{3,j,l} v_{j,\text{obs},l}$$

and

$$\lambda = (J_{\text{obj}} \Gamma^{-1} J_{\text{obj}}^T)^{-1} (\dot{x}_{\text{obj},d} - J_{\text{obj}} \Gamma^{-1} \Theta) \quad (25)$$

This new optimal joint velocity will again be solved repeatedly and a position-level optimum is iteratively updated inside the program cycle as explained in Section 3.2, whose final value will be used as the input to motion controller.

The proposed design makes use of the equality constraint to synchronise the workpiece motion intended by all manipulators, so the internal force problem due to potential conflicting motion is solved. The method is also novel in the sense that although each local optimisation problem is formulated based on joint-space situation and the optimum is a complete joint pose vector, the communicated data only contain a desired Cartesian pose of the grasped object. This makes it convenient to combine different types of robots for cooperation with better plug-and-play friendliness. This nice property is due to that each locally desired object pose implicitly reflects its joint-space demand, i.e., internal performance adjustment and whole-body obstacle avoidance, and it will be propagated to other agents during data exchange, so the resultant group decision will respect all local demands, even though the robots do not share vision on obstacles or explicit internal performance conditions.

In summary, an intuitive explanation of the whole design is: the local optima only regarding to the demand of each individual robot member are first computed and the corresponding object pose is broadcast for negotiation, leading to a global agreement that all robots must thereafter comply with while searching again for the locally best solution. It actually resembles how human negotiate. The pseudocode of the complete process is shown as Algorithm 2.

Algorithm 2 Coordinated optimisation

```

1: function GROUPOPTIM( $q_{\text{now}}, x_{d,\text{trp}}, p_{\text{obs}}$ )
2:    $q_{d0} := \text{LocalOptim}(q_{\text{now}}, x_{d,\text{trp}}, p_{\text{obs}})$ 
3:    $T_{\text{obj},d0} := f_{\text{FK}}(q_{d0})$ 
4:   Broadcast  $T_{\text{obj},d0}$  and  $g_{0,T,Ri}$ 
5:    $T_{\text{obj},d} := \text{mean}(\text{all } T_{\text{obj},d0,Ri}, \text{all } w_{Ri})$ 
6:   Initialise  $q_d := q_{\text{now}}, t_{\text{itr}} := 0$ 
7:   repeat
8:     Compute the object pose based on  $q_d$ 
9:     Compute key point positions based on  $q_d$ 
10:    Compute all Jacobians based on  $q_d$ 
11:    Compute  $\dot{x}_{\text{obj},d}$  and  $\dot{q}'_{\text{trp}}$ 
12:    Compute every  $v_{j,\text{obs},l}$  and  $W_{3,j,l}$ 
13:     $\dot{q}_d := \Gamma^{-1}(\Theta + J_{\text{obj}}^T \lambda)$ 
14:     $q_d := q_d + \dot{q}_d \Delta t_{\text{itr}}$ 
15:     $t_{\text{itr}} := t_{\text{itr}} + \Delta t_{\text{itr}}$ 
16:  until  $t_{\text{itr}} \geq T_{\text{itr}}$  ▷ or until  $\dot{q}_d \& \ddot{q}_d \approx 0$ 
17:  return  $q_d$ 
18: end function

```

5. Robot arm control

In order to enhance interaction safety, joint impedance control is utilised to shape the dynamic behaviour of the joints as a virtual mass–spring–damper with tunable parameters so that the robot reacts softly to physical contact and large interaction forces can be avoided. Since a precisely decoupled behaviour is unnecessary here, the version without inertia shaping is chosen, expressed as

$$\tau_m = K_P \tilde{q}_{Ri} - K_D \dot{q}_{Ri} + C_{Ri} \dot{q}_{Ri} + g_{Ri} - \tau_{f,Ri} - \tau_{e,Ri} \quad (26)$$

with

$$\tilde{q}_{Ri} = q_{d,Ri} - q_{Ri} \quad (27)$$

K_P and K_D are the stiffness and damping gain matrices, respectively. Substituting (26) into the robot dynamics (4) leads to

$$-M_{Ri} \ddot{q}_{Ri} - K_D \dot{q}_{Ri} + K_P \tilde{q}_{Ri} = \tau_{e,Ri} \quad (28)$$

characterising a joint-space impedance.

In fact, the algorithm proposed in Sections 3 and 4 can also be freely combined with other motion controllers. For example, Cartesian impedance control or even pure PID motion control is worth considering for potentially better Cartesian-space tracking performance. For the impedance controllers, [8] has shown some attractive stability properties in cooperative manipulation. The details will not be discussed here due to the scope of this paper.

6. Simulation

In order to evaluate the functionality of the proposed method under multi-obstacle environments, simulation in MATLAB Simulink with two Franka Emika Panda 7-DOF robot arms is performed and the results without and with negotiation will be compared. A video can be found online: <https://youtu.be/8N8ED3ZyEp4>. As shown in Fig. 3, the two robots are mounted “face to face” on the X axis of the world frame and there are 3 dynamic obstacles near them. Note that the simulation presented in this section is mostly qualitative since the data often contain the effects of multiple obstacles at the same time, making them difficult to interpret. A detailed and more comprehensive study will be provided later in the experiment section. The robot on the left is labelled as Robot 1 and the other robot being Robot 2. The distance between their bases is 0.956 m and the base frame of Robot 1 coincides with the world frame. This set-up is identical to a real experimental platform in the laboratory of the authors. Since it is hard to accurately simulate the stiffness and damping induced by the contact between robot fingers and a firmly grasped object, in this simulation only a 1-DOF linear spring–damper without mass is assumed to be connecting two grippers, and the grasping type is hinged. The internal force will therefore be studied only qualitatively here, too.

The initial joint pose of both robots is $[0, -\pi/4, 0, -3\pi/4, 0, \pi/2, \pi/4]^T$. Despite the hinged grasping in the simulated reality, the ideal control goal is still to accomplish the motion task while maintaining the rigid grasping formation. The initial pose of the virtual workpiece is chosen as the static desired transportation goal. Obstacle 1 as shown in Fig. 3 starts from (0.2, 0.5, 0.7) and moves along $-Y$ direction. Obstacle 2 starts from (0.45, -0.5 , 0.55) and Obstacle 3 starts from (1.3, -0.5 , 0.6), and both of them move along $+Y$ direction. With this set-up, Obstacle 1 will mainly interfere with the upper links of Robot 1, Obstacle 2 mainly with the workpiece, and Obstacle 3 with the elbow of Robot 2. Their moving speeds are all 0.05 m/s.

The internal performance cost contains two components — one for singularity avoidance, namely manipulability, and the other for joint limit avoidance. Its mathematical definition is

$$g_{\text{ic}} = g_{\text{ic},m} + g_{\text{ic},j} \quad (29)$$

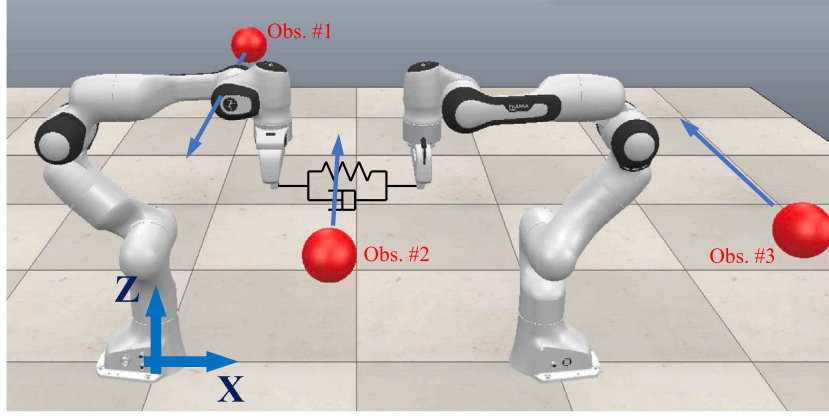


Fig. 3. The simulation setup using two Franka Emika Panda manipulators with 3 obstacles, rendered in CoppeliaSim.

with

$$\begin{aligned} g_{ic,m} &= \frac{1}{2} \cdot \frac{1}{m_{\max}^2} (m_{\max} - m(q))^2 \\ g_{ic,j} &= \frac{1}{2} \cdot \frac{4}{\|q_{\max} - q_{\min}\|^2} \cdot \|q - q_{\text{middle}}\|^2 \end{aligned} \quad (30)$$

where the subscript “Ri” is omitted. $m = \sqrt{\det(J_{\text{obj}} J_{\text{obj}}^T)}$ is the manipulability measure, and $m_{\max}$ is its maximum value (0.13 on Franka Emika Panda). $q_{\min}$, $q_{\max}$ and q_{middle} are the lower limit, upper limit and middle value of the joints, respectively. Since $g_{ic,m}$ and $g_{ic,j}$ have different dimensions, they are both scaled into the range [0, 0.5] for comparability. The internal cost optimising velocity $\dot{q}_{ic}$ is numerically evaluated online instead of using (10) and (11).

The weight for transportation is $w_{\text{trp}} = 1$, and the nominal weight for obstacle avoidance is $w_{\text{obs}} = 50$. The top-left 3-by-3 block of K_{trp} , i.e. the translational part, is equal to $10I$, and the bottom-right block is $20I$. For K_{glo} , the top-left block is I and the bottom-right $2I$. The null space velocity gain is $k_{ic} = 20$. The obstacle avoidance velocity gain is $k_{\text{obs}} = 2$. For the obstacle avoidance of the arm, Joint 4, Joint 6 and the end effector are chosen as the key points on the robot body. The reaction radius is $r_{\text{obs}} = 0.3$. The impedance control gains are $K_P = \text{diag}([60, 60, 60, 60, 40, 20, 10])$ and $K_D = \text{diag}([10, 10, 10, 8, 5, 2, 2])$. The stiffness of the spring connecting two grippers is 100 N/m and the damping is 100 N s/m.

The iterative solving process will not terminate according to the gradient. Instead, the number of loops is fixed by setting the iteration time step $\Delta t = 0.1$ s and a constant iteration time horizon $T_{\text{iter}} = 10$ s, leading to $101 \times 2 = 202$ total loops in each optimisation cycle. Also a fixed planning cycle time is set to be 0.1 s. Between each planning step, the low-level impedance controller running at 1 kHz (specified by robot manufacturer, Franka) will take the last updated reference pose. A first-order low-pass filter with a 0.5 time constant is implemented to smooth the output of the pose planner. The damping factor in the negotiation weight (19) is 1.0.

First the negotiation is turned off, i.e. Algorithm 1 is running on both robots without communication. As soon as the obstacles enter the reactive radius, the distortion of grasping formation due to different motion intentions can already be observed, as shown in Fig. 4.a1. When Obstacle 2 is crossing the middle area, it is obvious in Fig. 4.b1 that the spring-damper (illustrated in blue) is being pulled longer, because the two robots try to move into opposite directions according to their local optimisation results. After all obstacles passed the critical area, the end effectors go back to the desired Cartesian pose, and the null space optimisation is able to recover the preferred joint pose in sense of (29), which is close to the initial pose in this case, as shown in Fig. 4.c1.

When the coordinated motion planning, namely Algorithm 2, is used, it can be seen in Fig. 4.a2 that during the starting phase the

grasping formation is well maintained in spite of different local disturbances. Fig. 4.b2 shows that coordinated motion is also guaranteed when the obstacle is interfering with the workpiece and there is no visible deformation. In the end the robots can still recover the desired pose, as Fig. 4.c2 shows.

The internal force in the coupling spring-damper is calculated and depicted in Fig. 5, where the value is defined positive when the connection is being compressed. During the non-coordinated simulation, the spring-damper is mostly stretched due to the diverging local motion intentions and the maximum value reaches above 11N. Although the simulated connection is different from reality, it is still able to indicate a remarkable variation in the distance between two end effectors. When coordination is switched on, the overall internal load is significantly reduced to below 4N, where the spikes is a consequence of temporal fast change of the joint pose references. Since the nonlinear second-order behaviour (28) is subject to tracking delay and its corresponding Cartesian-space dynamics is inherently coupled, transient oscillation can happen when the optimisation output experiences a steep change. Nevertheless, the internal load data of this qualitative simulation show that the proposed method can effectively enhance the motion coordination during cooperative manipulation and thus reduce the conflicting forces and protect both the workpiece and robots. At the same time, the redundancy in the multi-robot system can be automatically utilised whenever possible for obstacle avoidance, as can be seen from the elbow motion of Robot 2.

7. Experiment

7.1. Set-up

A dual-arm experimental platform consisting of two Franka Emika Panda 7-DOF robot arms is available in the robotics laboratory of the Institute of Control Systems (LRS) in the University of Kaiserslautern, as shown in Fig. 6. The positioning of the manipulators is the same as in the simulation. Each robot is controlled by a laptop with Ubuntu 16.04. PREEMPT_RT patch is installed to meet real-time requirements. The control algorithm is implemented in C++ making use of the library provided by the robot manufacturer. The laptop for Robot 1 is powered by an Intel Core i7-3520M (2.9–3.6 GHz) and 8 GB RAM. The laptop for Robot 2 is powered by Intel Core 2 Duo CPU P8700 (2.53 GHz) and only 4 GB RAM.

A Hokuyo URG-04LX-UG01 2D laser scanner is placed on the table for obstacle detection in the X-Z plane. The robots will see the obstacles as lines with infinite length along the Y axis. Considering the hardware limitations in the lab and the fact that simulation in Section 6 has proven the effectiveness with multiple dynamic obstacles, only single-obstacle cases will be evaluated in the experiments, where a wooden

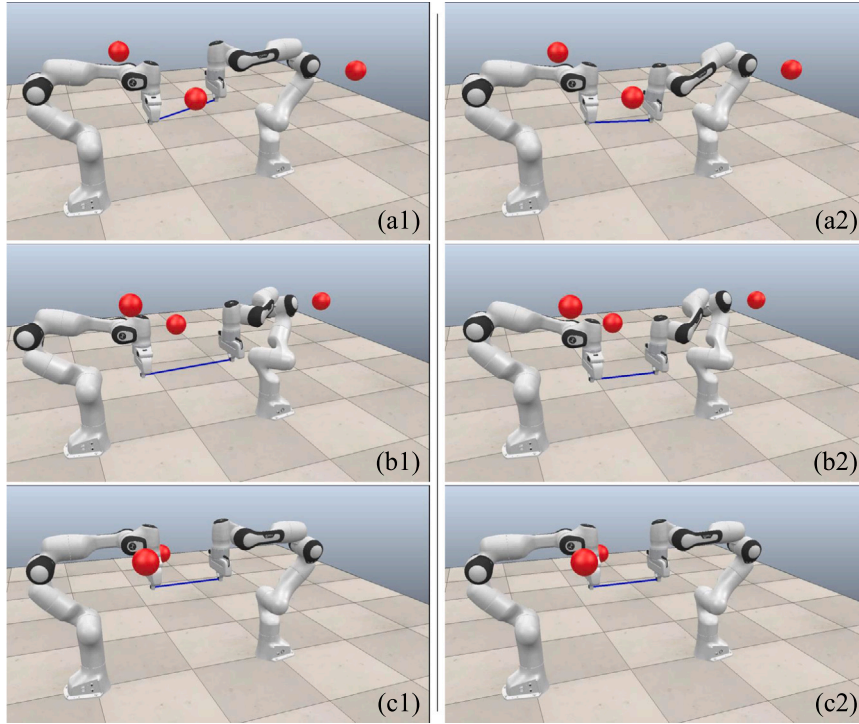


Fig. 4. Visualisation of the simulation without coordination as in (a1)-(b1)-(c1); and with coordination as in (a2)-(b2)-(c2). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

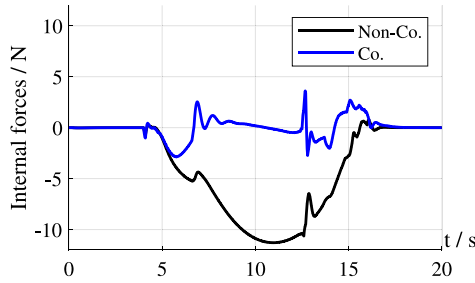


Fig. 5. The internal forces in the workpiece without and with the coordination mechanism.

stick will be used as the obstacle and it will always be placed in parallel with the world Y axis. The laser scanner data is processed by a laptop running real-time Ubuntu 16.04, powered by Intel Core i5-3210M (2.5–3.1 GHz) and 8 GB RAM. The detected obstacle positions will be transmitted to the two robots through wireless Ethernet LAN (WiFi). To simplify TCP/IP programming, this laptop also serves as a data exchanging server for the two robots, i.e. both robots communicate with this computer and $T_{obj,d}$ is actually computed here and then transferred back to each member. However it does not harm the distributed nature of the design, since this part of implementation could also be moved to individual robots. Additionally, although the information of all existing obstacles is accessible to every robot, each robot only reacts to those inside its reaction range, so the local sensing restriction still holds.

The same internal performance cost as in the simulation (29) is used, and the gradient (9) is still computed online to test the program speed in a harsh condition, although it is obvious that g_{ic} only depends on pose q and (11) could be adopted for computational efficiency. The initial robot poses and all control parameters are also designed identical to the simulation in Section 6.

Two sets of experiments are designed. In the first set, the transportation goal pose is constant, while in the second the transportation goal

is time-varying. In both sets a dynamic obstacle (the wooden stick) is brought into close proximity of the manipulation system by the human operator to test its responsiveness. A video of the experiments can be found online: <https://youtu.be/VtUhqSqThEI>.

7.2. Results with static target

In this part of the experiment, two different objects will be manipulated - a soft plastic tube and a rigid steel threaded rod, as shown in Fig. 7. Their gravity is ignored. The performance with and without the coordination strategy will be compared. As mentioned before, the transportation target pose is first set as static and equal to the initial object pose. The wooden stick will be held by the human operator and moved around the multi-robot system so that its obstacle avoidance behaviour can be observed.

First the soft tube is grasped, and the coordination mechanism is turned off, therefore Algorithm 1 is running on each robot and there is no data exchange between the two robots. The obstacle is placed to four different positions in the X-Z plane, as shown in Fig. 8 in red circles. In the first situation, the optimal motion planner on Robot 1 has automatically exploited the redundancy to generate a pure internal joint motion to avoid the stick without changing the end effector pose in Cartesian space. After the obstacle is moved out of the reaction range, the internal performance optimisation component will drive the robot arm back to the originally desired pose. In the second situation shown in the top-right of Fig. 8, the obstacle is approaching the grasped object and both robots are trying to avoid it. However due to the lack of negotiation, they move in opposite directions and the object is obviously deformed. The bottom-left sub-figure shows that the same algorithm is implemented on both manipulators. In the last situation, the obstacle is only close to Robot 2, so Robot 1 has not performed any motion while its teammate is responding to the surroundings. As a consequence, large deformation has happened on the object.

The same routine is repeated with coordination turned on, and the experiment can be seen in Fig. 9. In the two sub-figures on the left, there are no noticeable difference with the previous test, since no Cartesian

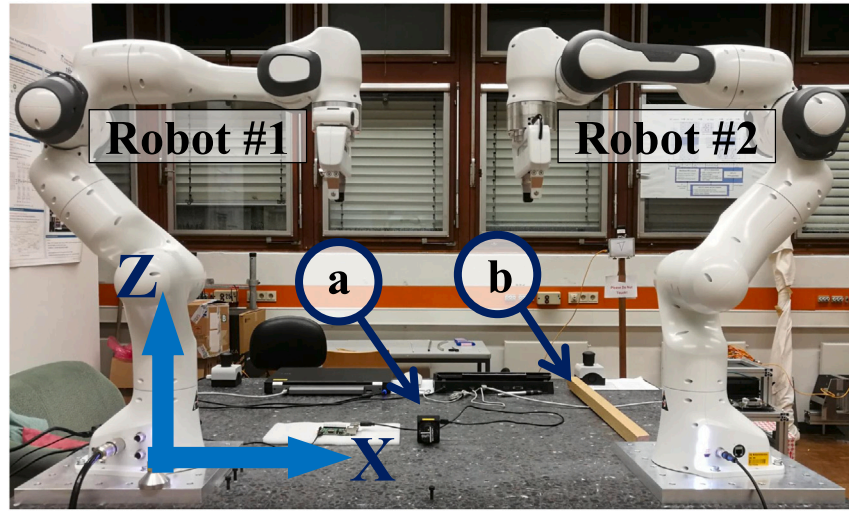


Fig. 6. The experimental setup in the robotics lab of LRS [28], where (a) is a 2D laser scanner and (b) is a wooden stick as an obstacle.



Fig. 7. The initial grasping states of the two objects, the elastic tube on the left and the steel threaded rod on the right.

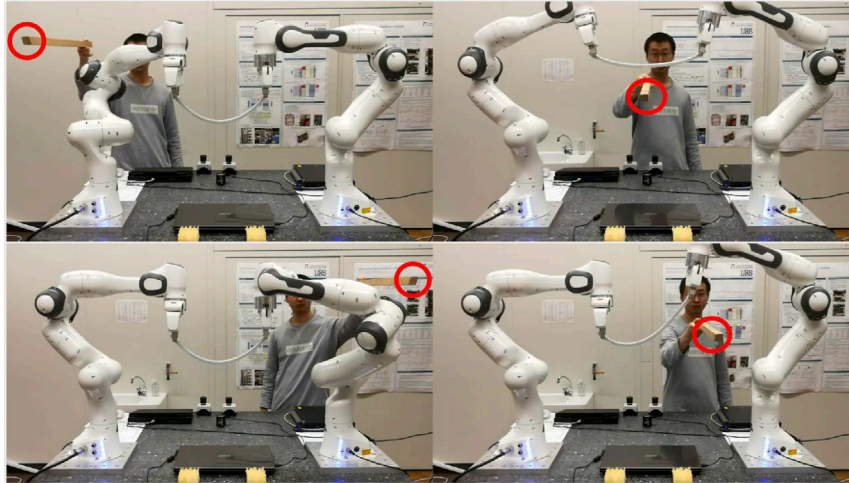


Fig. 8. The robot poses during the experiment without coordination — elastic object and static transportation goal.

motion is required and in this case the coordination mechanism plays no role. In the top-right scene however two manipulators can move upwards while maintaining the grasping formation, and there exists no visible deformation. The plastic tube looks very close to the obstacle because its curved shape is not included in this program, and the robots see the workpiece as a straight rod connecting the two grippers. In the last picture in Fig. 9, although the obstacle is only visible to Robot 2, its demand is implicitly transmitted to Robot 1 and now two robots are cooperating in the avoidance motion. Despite the orientation change, the intended object motion is synchronised and the tube is not deformed. Another interesting observation in this situation is that

Robot 2 is not moving as much as in the previous test, because while Robot 1 is catering to Robot 2, its local cost also increases and it “wants” to go back. This intention will be reflected in the negotiation result and Robot 2 has to respect it during the second round of iterative quadratic programming, therefore it is not forcing the other member to comply with itself. In short, the resulting coordinated motion is considerate of all agents in the cooperation group.

During the experiment of the coordinated motion, the planned cost in the last loop of the first iterative quadratic programming is recorded and plotted in Fig. 10. A trade-off between transportation goal tracking and obstacle avoidance appears every time when the wooden stick is

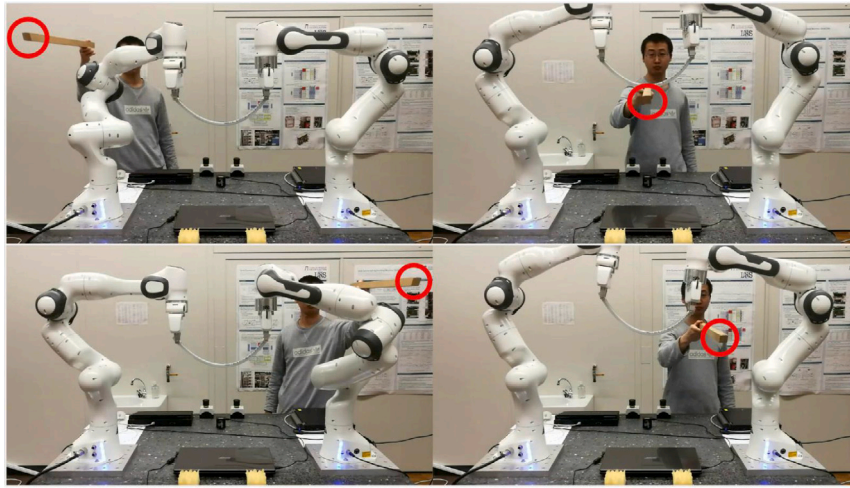


Fig. 9. The robot poses during the experiment with coordination — elastic object and static transportation goal.

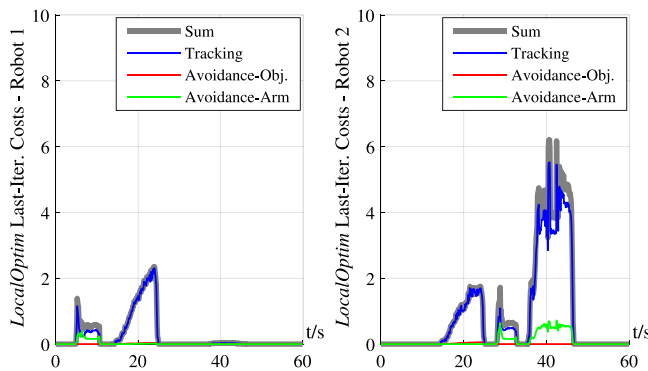


Fig. 10. The last-step cost during local optimisation with coordination turned on — elastic object and static transportation goal.

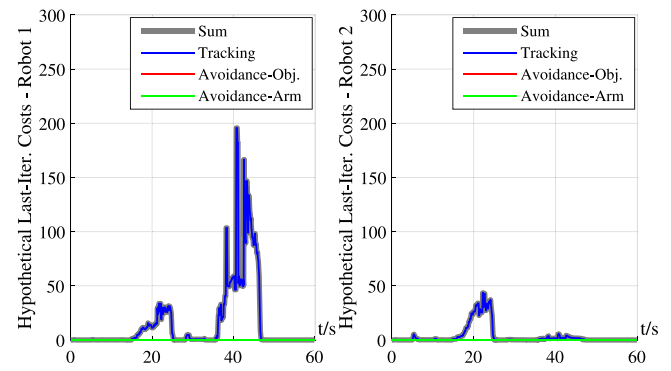


Fig. 11. The last-step cost during the hypothetical constrained optimisation using the first cost function — elastic object and static transportation goal.

placed close to the robots or the object. Since the obstacle avoidance weight is 50 times larger than the tracking weight, more emphasis is put on the former and its cost is usually lower in the run-time.

After the test, the recorded motion data and obstacle data are used for running a simulated iterative quadratic programming process with the cost function in the first optimisation problem (6), yet constrained by the recorded $T_{obj,d}$. The purpose is to compare the costs with and without the global agreement constraint based on the same cost function. The last-step cost is depicted in Fig. 11. As can be seen, the values are much higher than those in Fig. 10, because the robot members have influence on each other. The physical connection makes it necessary to sacrifice some local optimality to avoid conflicting motion and forces. The real last-step cost based on the second optimisation problem (20) is shown in Fig. 12.

The workpiece is then replaced with the steel rod, and the same two tests are conducted with the same obstacle positioning. Due to the rigidity of the object, hardly any visible deformation may happen, but the high stiffness makes internal forces in the system sensitive to motion conflict. The Franka Emika Panda robot is equipped with joint torque sensors and an estimation of the end effector wrench can be returned from its built-in control interface, which are used to calculate the internal forces based on (3). When there is no communication, the internal wrenches are as shown in Fig. 13. Those with coordination turned on are shown in Fig. 14.

Obvious difference can be seen from the two figures. When the human operator positions the wooden stick below the rod, similar to the very first experiment the robots tend to move in different

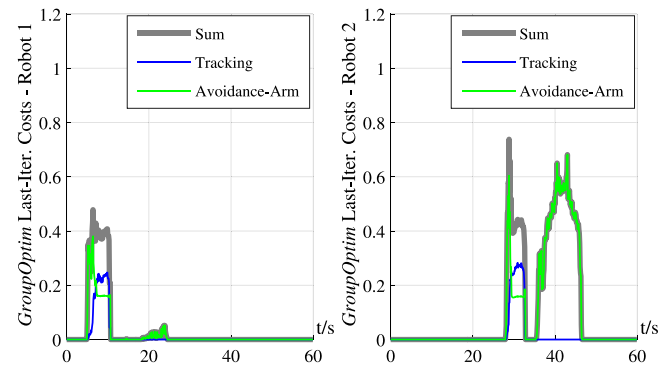


Fig. 12. The real last-step cost during the final optimisation — elastic object and static transportation goal.

directions and a large internal load in the X-Z plane arises, however when Algorithm 2 is used, this internal load vanishes. Note that in the fourth obstacle situation, the internal forces without coordination should have been much larger, however due to the structure of the end effector fingers, sliding happens consistently at the grasping point and the actual interaction forces are small. As shown on the left side of Fig. 15, without negotiation the angle between the rod and the gripper has already substantially changed, potentially leading to high internal load if the grasping is firm. Such phenomenon is not happening in the other sub-figure. Moreover, the tracking delay of the impedance control can cause some un-coordinated motion in transient, which is the reason

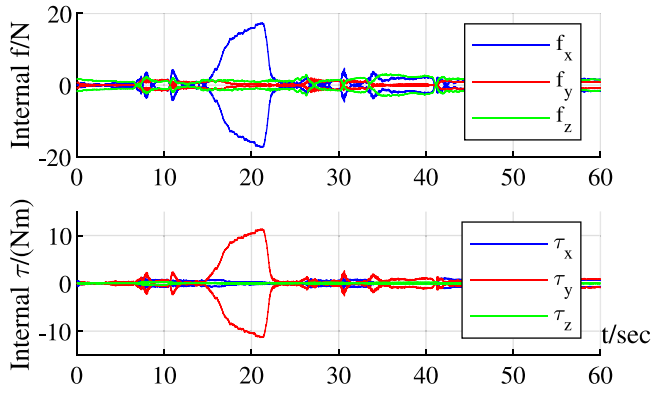


Fig. 13. The internal forces during the experiment with coordination turned off — rigid object and static transportation goal.

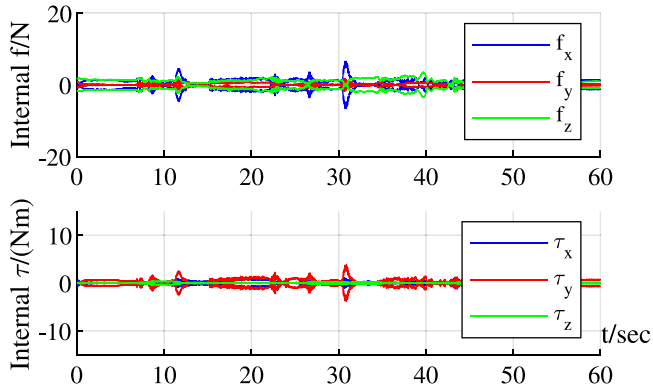


Fig. 14. The internal forces during the experiment with coordination turned on — rigid object and static transportation goal.

for the tiny spikes in the curve. It could be improved by using a faster controller or designing slower motion for the robot.

Since the grasping formation is always well maintained by the coordinative motion planner and the obstacle has been moved to the same positions, the robot motion is almost the same as in the experiment with the soft object. Thus the cost data are also nearly identical to Figs. 10, 11 and 12, and for spatial consideration they will not be shown here.

7.3. Results with moving target

To demonstrate the application of the proposed method to dynamic transportation, a simple experiment with time-varying transportation goal is designed with the steel rod as the workpiece. Starting from the initial pose, the desired object position alters every 3 s between ± 0.1 m in Z direction. As shown in the first row of Fig. 16, without the interference of an obstacle, the proposed optimisation strategy

can guide the robot to perfectly track the given trajectory. When the obstacle is placed near the elbow of the robot, the manipulation system is still able to accomplish the transportation task while avoiding the wooden stick using null space motion, as the second row shows. Then the obstacle is put below the workpiece when it is trying to move downwards, and above it when it goes up. By comparing the last row in Fig. 16 with the first row, it is evident that the presented algorithm is able to successfully restrict object motion to avoid collision also during dynamic goal tracking, making it applicable to real transportation and manipulation scenarios.

The internal wrenches are shown in Fig. 17. Only small oscillations can be seen on the data curves when the desired object pose changes and the multi-robot system is in a transient phase. Obstacle avoidance motion has not caused visible internal load.

7.4. Computation time

In all the presented experiments, on Robot 1 the computation time of coordinated optimal pose generator, i.e. Algorithm 2 with 202 iteration loops in total, is in the range of [4.9, 5.561] (ms). On the Robot 2 it is [11.425, 12.115] (ms). In fact both computers used in these experiments are already somewhat outdated. On a modern system the program could definitely run even faster, and a little tuning on the iteration parameters or using (11) instead of (9) could also help to further improve the planning speed. As a comparison, many existing collision-free motion planning algorithms are designed for single manipulators and they still spend from tens of milliseconds to a few seconds for each planning step, since they attempt to compute a whole trajectory [11–13]. Meanwhile, compared to the low-layer control cycle time of commercial robot products, e.g. ABB IRC5 controller working at 125 Hz and UR robots at 500 Hz, the computation time of the proposed algorithm on a modern computer being under 10 ms can be considered quite adequate for real-time applications.

8. Conclusion

This paper has introduced a novel distributed optimal control strategy for multi-robot manipulation systems. Not only the ultimate manipulation and transportation task can be accomplished, but also dynamic obstacle avoidance is effectively tackled. Additionally, to make better use of the reduced workspace of each robot arm, internal performance improvement is also an objective of the proposed control. To avoid directly solving a nonlinear position-level optimisation problem, a quadratic programming problem at velocity level is formulated, whose analytical solution is derived. A fast iterative solving process is then utilised to obtain a position-level reference. In order to guarantee synchronised motion and get rid of internal forces, a plug-and-play-friendly coordination mechanism is designed based on the aforementioned optimisation algorithm. By negotiation, the robots will implicitly take the demand of each other into consideration when determining the final optimal motion, and in the end no conflicting forces will be generated. The algorithm is tested in experiments on a dual-arm platform made up of two redundant manipulators. Results have shown

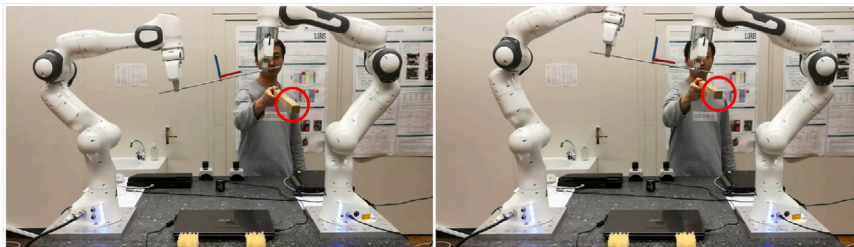


Fig. 15. The robot poses during the experiment with rigid object and static transportation goal. Left: Coordination off; Right: Coordination on.

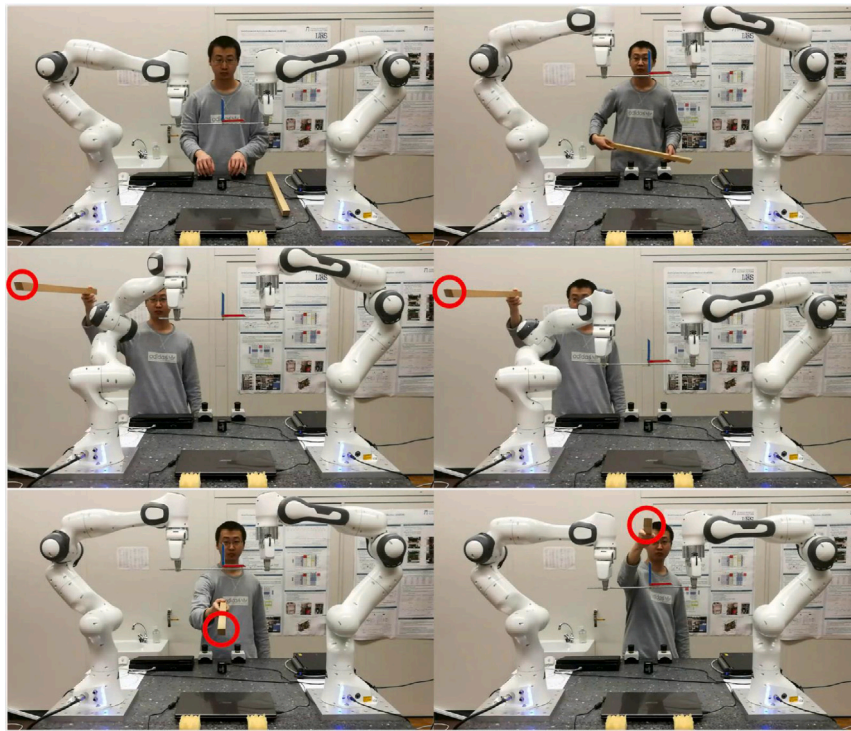


Fig. 16. The robot poses during the experiment with a rigid object and moving transportation goal.

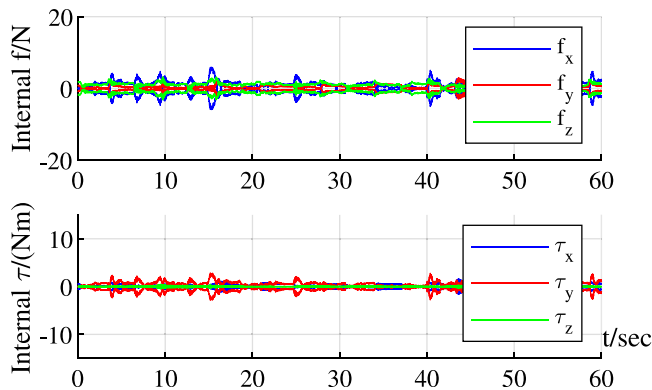


Fig. 17. The internal forces during the experiment with a rigid object and moving transportation goal.

that the presented control scheme can automatically utilise the redundancy of the system for obstacle avoidance, and the predefined grasping geometry can be maintained to avoid internal load. Its application to dynamic and responsive trajectory tracking is also demonstrated. The computation is fast, costing only a few milliseconds per planning step on an up-to-date computer.

CRedit authorship contribution statement

Yanhao He: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Min Wu:** Conceptualization, Resources, Writing - review & editing. **Steven Liu:** Conceptualization, Methodology, Resources, Writing - original draft, Writing - review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

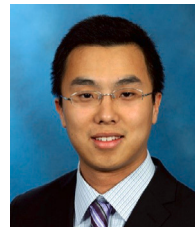
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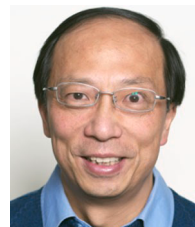
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